

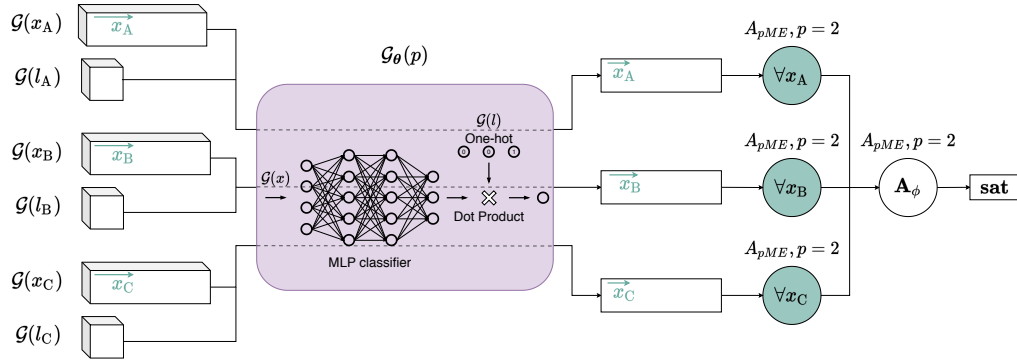


Figure 10: Symbolic Tensor Computational Graph for the Multi-Class Single-Label Problem. As before, the dotted lines in the figure indicate the propagation of activation from each input through the network, in this case producing three outputs.

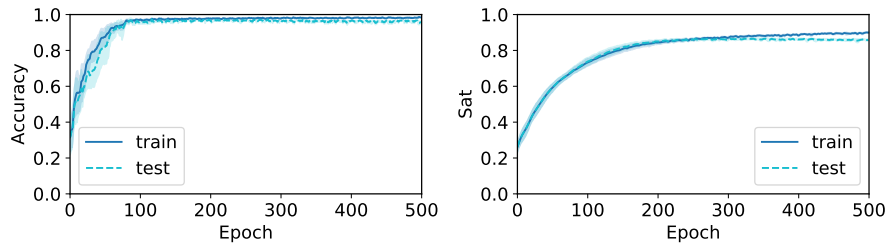


Figure 11: Multi-Class Single-Label Classification: Classification accuracy (left) and satisfaction level (right).

to each other, thus becoming a powerful tool in the case of the multi-label problem when typically the labeled data is scarce. We explore the *Leptograpsus crabs* data set [10] consisting of 200 examples of 5 morphological measurements of 50 crabs. The task is to classify the crabs according to their color and sex. There are four labels: blue, orange, male, and female. The color labels are mutually exclusive, and so are the labels for sex. LTN will be used to specify such information logically.

Domains:

items denoting the examples from the crabs dataset.
 labels denoting the class labels.

Variables:

$x_{\text{blue}}, x_{\text{orange}}, x_{\text{male}}, x_{\text{female}}$ for the positive examples of each class.
 x , used to denote all the examples.
 $\mathbf{D}(x_{\text{blue}}) = \mathbf{D}(x_{\text{orange}}) = \mathbf{D}(x_{\text{male}}) = \mathbf{D}(x_{\text{female}}) = \mathbf{D}(x) = \text{items}.$

Constants:

$l_{\text{blue}}, l_{\text{orange}}, l_{\text{male}}, l_{\text{female}}$ (the labels for each class).
 $\mathbf{D}(l_{\text{blue}}) = \mathbf{D}(l_{\text{orange}}) = \mathbf{D}(l_{\text{male}}) = \mathbf{D}(l_{\text{female}}) = \text{labels}.$

Predicates:

$P(x, l)$, denotes the fact that item x is labelled as l .
 $\mathbf{D}_{\text{in}}(P) = \text{items, labels}.$

Axioms:

$$\forall x_{\text{blue}} P(x_{\text{blue}}, l_{\text{blue}}) \quad (30)$$

$$\forall x_{\text{orange}} P(x_{\text{orange}}, l_{\text{orange}}) \quad (31)$$

$$\forall x_{\text{male}} P(x_{\text{male}}, l_{\text{male}}) \quad (32)$$

$$\forall x_{\text{female}} P(x_{\text{female}}, l_{\text{female}}) \quad (33)$$

$$\forall x \neg(P(x, l_{\text{blue}}) \wedge P(x, l_{\text{orange}})) \quad (34)$$

$$\forall x \neg(P(x, l_{\text{male}}) \wedge P(x, l_{\text{female}})) \quad (35)$$

Notice how logical rules 34 and 35 above represent the mutual exclusion of the labels on colour and sex, respectively. As a result, negative examples are not used explicitly in this specification.

Grounding:

$\mathcal{G}(\text{items}) = \mathbb{R}^5$; the examples from the data set are described using 5 features.
 $\mathcal{G}(\text{labels}) = \mathbb{N}^4$; one-hot vectors are used to represent class labels.²¹

$\mathcal{G}(x_{\text{blue}}) \in \mathbb{R}^{m_1 \times 5}$, $\mathcal{G}(x_{\text{orange}}) \in \mathbb{R}^{m_2 \times 5}$, $\mathcal{G}(x_{\text{male}}) \in \mathbb{R}^{m_3 \times 5}$, $\mathcal{G}(x_{\text{female}}) \in \mathbb{R}^{m_4 \times 5}$. These sequences are not mutually-exclusive, one example can for instance be in both x_{blue} and x_{male} .
 $\mathcal{G}(l_{\text{blue}}) = [1, 0, 0, 0]$, $\mathcal{G}(l_{\text{orange}}) = [0, 1, 0, 0]$, $\mathcal{G}(l_{\text{male}}) = [0, 0, 1, 0]$, $\mathcal{G}(l_{\text{female}}) = [0, 0, 0, 1]$.

²¹There are two possible approaches here: either each item is labeled with one multi-hot encoding or each item is labeled with several one-hot encodings. The latter approach was used in this example.

$\mathcal{G}(P \mid \theta) : x, l \mapsto l^\top \cdot \text{sigmoid}(\text{MLP}_\theta(x))$, with the MLP having four output neurons corresponding to as many classes. As before, $\cdot$ denotes the dot product which selects a single output. By contrast with the previous example, notice the use of a `sigmoid` function instead of a `softmax` function.

Learning:

As before, the fuzzy logic operators and connectives are approximated using the stable product configuration with $p = 2$ for A_{pME} , and for the formula aggregator, A_{pME} is also used with $p = 2$.

Figure 12 shows the result of the Adam optimizer using backpropagation trained with batches of 64 examples. This time, the accuracy is defined as $1 - \text{HL}$, where HL is the average Hamming loss, i.e. the fraction of labels predicted incorrectly, with a classification threshold of 0.5 (given an example u , if the model outputs a value greater than 0.5 for class C then u is deemed as belonging to class C). The rightmost graph in Figure 12 illustrates how LTN learns the constraint that a crab cannot have both blue and orange color, which is discussed in more detail in what follows.

Querying:

To illustrate the learning of constraints by LTN, we have queried three formulas that were not explicitly part of the knowledge-base, over time during learning:

$$\phi_1 : \quad \forall x (P(x, l_{\text{blue}}) \rightarrow \neg P(x, l_{\text{orange}})) \quad (36)$$

$$\phi_2 : \quad \forall x (P(x, l_{\text{blue}}) \rightarrow P(x, l_{\text{orange}})) \quad (37)$$

$$\phi_3 : \quad \forall x (P(x, l_{\text{blue}}) \rightarrow P(x, l_{\text{male}})) \quad (38)$$

For querying, we use $p = 5$ when approximating the universal quantifiers with A_{pME} . A higher p denotes a stricter universal quantification with a stronger focus on outliers (see Section 2.4).²² We should expect ϕ_1 to hold true (every blue crab cannot be orange and vice-versa²³, and we should expect ϕ_2 (every blue crab is also orange) and ϕ_3 (every blue crab is male) to be false. The results are reported in the rightmost plot of Figure 12. Prior to training, the truth-values of ϕ_1 to ϕ_3 are non-informative. During training one can see, with the maximization of the satisfaction of the knowledge-base, a trend towards the satisfaction of ϕ_1 , and an opposite trend of ϕ_2 and ϕ_3 towards *false*.

4.4. Semi-Supervised Pattern recognition

Let us now explore two, more elaborate, classification tasks, which showcase the benefit of using logical reasoning alongside machine learning. With these two examples, we also aim to provide a more direct comparison with a related neurosymbolic system DeepProbLog [41]. The benchmark examples below were introduced in the DeepProbLog paper [41].

²²Training should usually not focus on outliers, as optimizers would struggle to generalize and tend to get stuck in local minima. However, when querying ϕ_1, ϕ_2, ϕ_3 , we wish to be more careful about the interpretation of our statement. See also 3.1.3.

²³Notice how, strictly speaking, other colours remain possible since the prior knowledge did not specify the bi-conditional: $\forall x (P(x, l_{\text{blue}}) \leftrightarrow \neg P(x, l_{\text{orange}}))$